

Macromolecular Boron-Based Salt Enables Dense Interphases for Long-Cycling Lithium-Sulfur Batteries

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Lithium-sulfur (Li-S) batteries represent a compelling next-generation energy storage system with practical energy densities exceeding 700 Wh kg^{-1} , offering a promising pathway beyond current lithium-ion technology. However, their commercial viability remains constrained by deleterious interfacial reactions between lithium metal anodes and polysulfide-containing electrolytes. Herein, it is presented a molecular engineering approach through a novel boron-based salt, lithium perfluoropinacolatoborate (LiFPB), strategically designed to reinforce the solid electrolyte interphase (SEI) for long-cycling Li-S batteries. LiFPB anions, featuring higher specific charge (mass-to-charge ratio) and larger steric bulk compared to conventional salts, demonstrate enhanced resistance to Helmholtz double-layer repulsion and increased susceptibility to lithium metal reduction, promoting the formation of a robust SEI enriched with LiF and LiB_xO_y species. The LiFPB-containing electrolyte exhibits superior lithium metal compatibility, achieving a high coulombic efficiency of 99.59%. Consequently, Li-S cells demonstrate markedly improved capacity retention from 50.9% to 75.7% over 200 cycles. This strategy has been successfully scaled to Ah-level Li-S pouch cells, achieving practical energy densities of 408 Wh kg^{-1} with stable cycling over 75 cycles. This work presents an effective approach to developing long-cycling Li-S batteries through the rational design of electrolyte salt.

1. Introduction

Lithium-sulfur (Li-S) batteries have great potential to achieve practical energy densities exceeding 700 Wh kg^{-1} , a crucial threshold for enabling heavy-duty vehicles and electric flight.^[1,2] However, the intrinsic precipitation-dissolution conversion mechanism of the S cathode generates soluble lithium polysulfides (LiPSs) as essential intermediates, which inevitably trigger the notorious shuttle effect and subsequent parasitic reactions between lithium (Li) metal anodes and LiPSs. Under practical cell design conditions (high sulfur loading, lean electrolyte, and ultrathin Li metal), the accelerated electrolyte degradation, primarily stemming from these parasitic reactions at the Li metal surface, fundamentally degrades the cycling stability of Li-S batteries.^[3,4] Therefore, designing a robust solid-electrolyte interphase (SEI) to stabilize Li metal anodes against LiPS-induced deterioration plays a crucial role in realizing long-cycling Li-S batteries.^[5,6]

The electrolyte composition and its derived SEI fundamentally govern the cycling stability of Li-S batteries. Among various electrolyte components, lithium salts play a pivotal role in enhancing the coulombic efficiency (CE) of Li-S batteries by facilitating the formation of dense, insoluble inorganic components within the SEI layer. Notably, the strategic incorporation of LiNO_3 has demonstrated remarkable effectiveness in suppressing the polysulfide shuttle effect, enabling Li-S batteries to achieve CEs exceeding 99%.^[7] However, the limited solubility of LiNO_3 and its progressive decomposition during cycling compromise the long-term cycling performance stability of Li-S batteries, particularly under low electrolyte-to-sulfur (E/S) ratios.^[8] Lithium bis(fluorosulfonyl)imide (LiFSI) and lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) demonstrate enhanced compatibility with Li metal anodes due to the formation of LiF-rich SEI. Nevertheless, the S=O structure in LiTFSI molecules acts as an electrophilic center susceptible to nucleophilic attack from terminal sulfide species (S^-) present in lithium polysulfides (Li_2S_n).^[9] This mechanistic understanding explains why many LiF-forming sulfonate-based lithium salts prove incompatible as electrolyte components in Li-S batteries. Therefore, the rational design of novel lithium salts that can

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facilitate a robust SEI while maintaining chemical stability with Li_2S_n emerges as a critical challenge for advancing stable Li-S batteries.

During lithium plating, the significant volumetric expansion of the Li metal anode leads to continuous SEI rupture, necessitating persistent SEI reformation. Electric double layer plays a crucial role in SEI formation. At the anode/electrolyte interface, a Li^+ -rich but anion-depleted inner-Helmholtz layer spontaneously forms through the self-assembly of solvent molecules during charging. This interfacial structure fundamentally dictates the result of interphasial chemistry; notably, the negatively charged electrode surface electrostatically repels anions from the inner Helmholtz layer, significantly impeding the formation of critical inorganic SEI components.^[10] To address these challenges, various electrolyte engineering strategies have been developed to establish favorable interphases, including high-concentration electrolytes (HCEs)^[11] and localized high-concentration electrolytes (LHCEs).^[12] However, these approaches often compromise sulfur utilization and adversely affect the discharge voltage plateau. Theoretically, anions with higher specific charge (mass-to-charge ratio) exhibit reduced sensitivity to applied electric fields, thereby demonstrating enhanced resistance to electric double-layer repulsion. This characteristic potentially increases the probability of anion reduction and subsequent formation of desirable inorganic SEI components during charging.

In this work, we present an alternative approach to stabilizing Li metal interfaces, diverging from conventional HCE or LHCE strategies. We rationally designed and synthesized a novel lithium perfluoropinacolatoborate (LiFPB) salt through a single-step reaction in anhydrous conditions, strategically incorporating both C-F and B-containing functionalities. The as-synthesized LiFPB demonstrates dual advantages: exceptional chemical stability with soluble polysulfides and superior SEI formation. Notably, LiFPB (682 g mol^{-1}) possesses a significantly higher specific charge ($682/280$) (2.43 times that of TFSI^-) and a substantially lower LUMO energy level, facilitating the formation of a robust SEI enriched with LiF and LiB_xO_y species. The LiFPB-containing electrolyte exhibits remarkable compatibility with Li metal anodes, achieving an unprecedented coulombic efficiency of 99.59%. This superior interfacial stability translates into exceptional performance in practical applications: an Ah-level Li-S pouch cell employing the LiFPB-containing electrolyte delivers a practical energy density of 408 Wh kg^{-1} while maintaining stable cycling performance over 75 cycles. This novel lithium salt paves the way for boosting the development of practical Li-S batteries.

2. Results and Discussion

The electrostatic force experienced by anions near the electrode surface follows Coulomb's law ($F = qE$, where F represents force, q denotes charge, and E indicates electric field strength). Consequently, anions possessing higher specific charge (mass-to-charge ratio) demonstrate diminished mobility in response to applied electric fields. These sterically hindered or high molecular weight anions exhibit enhanced resistance to electrostatic repulsion from the Helmholtz double-layer during lithium plating processes. This reduced mobility enables them to remain in closer proximity to the electrode surface, fundamentally altering the composition and morphology of the resulting SEI layer

and subsequently influencing lithium deposition behavior. The LiFPB salt with a molecular weight of 682 g mol^{-1} (Figure 1a) was synthesized via a one-pot reaction of lithium borohydride (LiBH_4) with perfluoropinacol in a molar ratio of 1:2, using tetrahydrofuran (THF) as the solvent at 60°C (Figure S1, Supporting Information). This approach minimizes the introduction of impurities, as the entire process is conducted in an organic solvent environment, thereby reducing the need for additional purification steps.^[13,14] The peak observed at $m/z = 674.95$ in the mass spectrometry (MS) analysis (Figure 1b) aligns precisely with the theoretical molecular weight of the FPB^- anion. Nuclear magnetic resonance (NMR) analysis further validates the structural integrity and purity of the compound. A single distinct peak was observed in the ^{11}B NMR spectrum at 11.40 ppm (Figure 1c) and in the ^{19}F NMR spectrum at -69.24 ppm (Figure 1d), with no additional peak indicative of impurities. These results confirm the presence of only one type of boron and fluorine chemical environment within the salt, underscoring its high purity and structural homogeneity. Upon incorporating 0.2 M LiFPB, 0.8 M LiTFSI, and 1 wt.% LiNO_3 into a solution comprising DME and DOL at a volumetric ratio of 1:1, the ^{11}B NMR spectrum retains its characteristic peak at 11.40 ppm (Figure S2a, Supporting Information), while the ^{19}F NMR spectrum exhibits two peaks at -69.24 and -79.80 ppm (Figure S2b, Supporting Information), the latter peak attributed to LiTFSI.^[15] In the ^{13}C NMR spectrum, two distinct sets of carbon signals are observed, corresponding to the C=O and $-\text{CF}_3$ groups in the FPB^- anion (Figure S3, Supporting Information). We conducted a series of reactions between polysulfide solutions and both LiFPB and LiTFSI (Figure S4, Supporting Information). Upon analysis of the mixture containing LiTFSI and Li_2S_6 , a notable shift in the characteristic ^{19}F NMR of LiTFSI was observed, migrating from -79.8 to -79.52 ppm (Figure S5a, Supporting Information). Concurrently, a distinct by-product emerged at -75.39 ppm (Figure S5b, Supporting Information). These findings unequivocally demonstrate the inherent incompatibility between LiTFSI and polysulfides, highlighting the susceptibility of LiTFSI to nucleophilic attack by polysulfide species. In stark contrast, when LiFPB was combined with Li_2S_6 , the characteristic ^{19}F NMR peak of LiFPB exhibited remarkable stability, remaining invariant. This observation underscores the exceptional resistance of LiFPB to nucleophilic attack by polysulfides, thereby establishing its superior compatibility with these compounds (Figure S6, Supporting Information). The low reactivity may be due to the high steric hindrance of high fluorine groups and the low valence state of boron (+3). This unique structural arrangement imparts LiFPB an extraordinary degree of chemical robustness, rendering it impervious to the deleterious effects of polysulfide interactions. These results indicate that LiFPB exhibits excellent chemical compatibility with polysulfides. The electronic properties of the synthesized FPB^- anion were characterized through frontier molecular orbital analysis, which are critical parameters for evaluating the SEI-forming ability. Notably, the FPB^- anion (4.55 eV) exhibits a significantly lower LUMO energy level compared to the TFSI^- anion (4.75 eV) (Figure 1e), indicating enhanced thermodynamic favorability toward electrochemical reduction and subsequent SEI formation. Density functional theory (DFT) calculations further elucidated the interfacial interaction mechanisms between LiFPB and lithium metal. The calculated binding energy for LiFPB

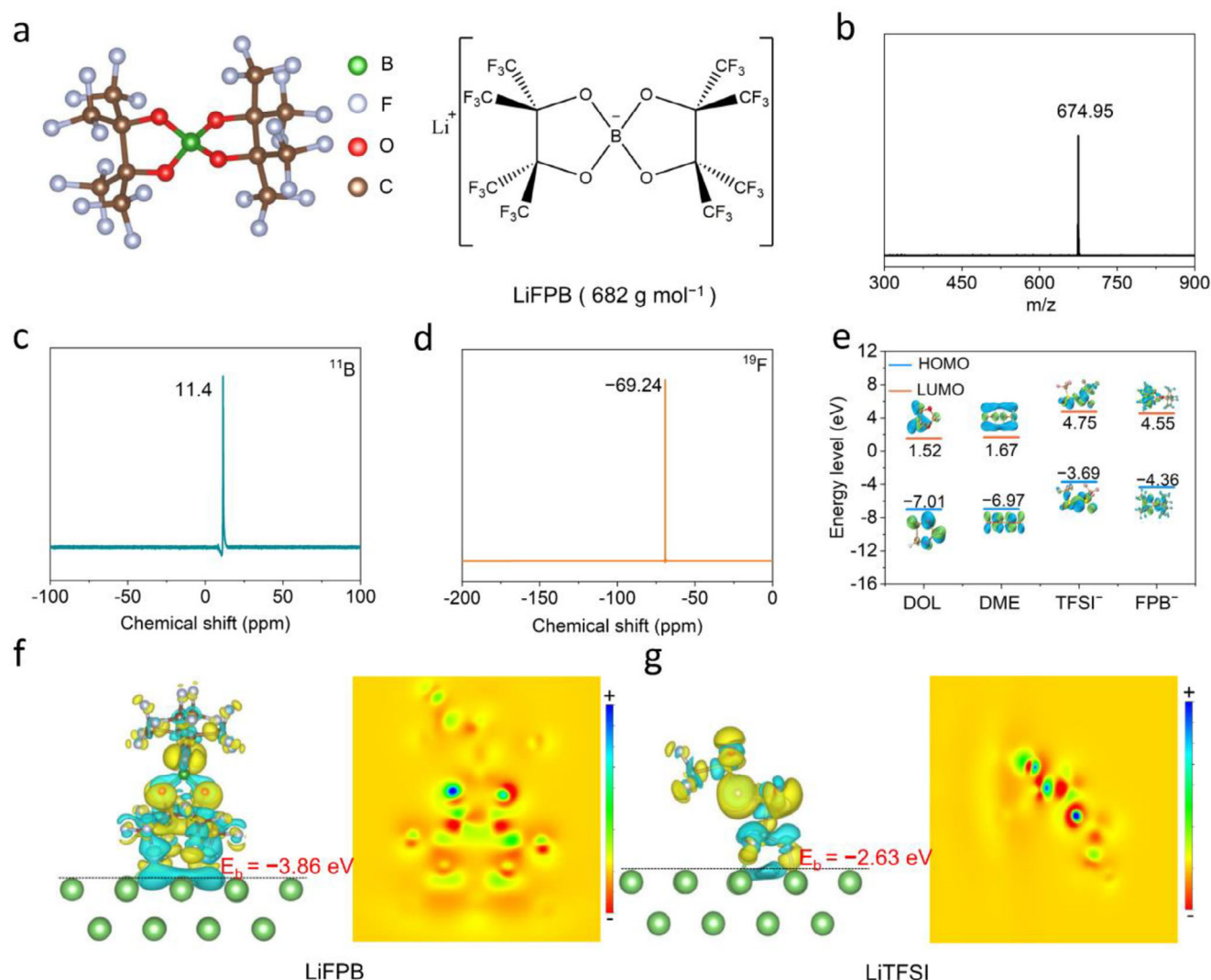


Figure 1. a) Chemical structures of LiFPB. b) Mass spectrometry of anion species of LiFPB. c) ^{11}B NMR and d) ^{19}F NMR spectra of LiFPB in deuterated dimethyl sulfoxide (DMSO- d_6). e) Calculated HOMO and LUMO energy levels of DOL, DME, TFSI $^-$ anion, and FPB $^-$ anion. f, g) The atomic configurations of LiFPB and LiTFSI adsorbed on the Li surface, along with their corresponding binding energies and differential charge density distributions.

adsorption on the lithium surface (-3.86 eV , Figure 1f) substantially exceeds that of LiTFSI (-2.63 eV , Figure 1g), providing compelling evidence that the fluorine moieties in LiFPB more readily participate in LiF formation at the lithium metal interface.^[15]

The electrochemical performance of the LiFPB salt was systematically evaluated through Li||Cu half-cells and Li||Li symmetric cell configurations. In Li||Li symmetric cells, the LiFPB-containing electrolyte demonstrated remarkable interfacial stability, exhibiting consistently low and uniform polarization voltage profiles throughout extended cycling exceeding 500 h (Figure 2a). In contrast, symmetric cells employing conventional electrolyte formulations displayed progressively increasing polarization voltages after 380 h (Figure 2b), indicative of substantial interfacial resistance accumulation.^[16] The cycling stability of Li metal was further assessed using Li||Cu asymmetric cells. Under controlled electrodeposition conditions (1 mAh cm^{-2} capacity at 0.5 mA cm^{-2} current density), cells with conventional elec-

trolyte maintained $>90\%$ coulombic efficiency (CE) for merely 35 cycles, subsequently deteriorating to below 65% after 100 cycles (Figure 2c,e). Conversely, cells incorporating LiFPB-containing electrolyte sustained exceptional CE ($>98.5\%$) throughout 100 cycles. Cumulative analysis of irreversible lithium consumption over 100 cycles demonstrated that conventional electrolyte formulations depleted 14.74-fold ($14.89 \text{ mAh/cm}^2/1.01 \text{ mAh/cm}^2$) more lithium metal compared to the LiFPB-containing electrolyte (Figure 2d), demonstrating the superior interfacial compatibility of LiFPB with Li metal anodes.^[17]

Aurbach's measurement protocol showed that the Li||Cu cell with LiFPB-containing electrolyte achieved a CE of 99.59%, significantly outperforming the conventional electrolyte (97.73%) (Figure 2f). This result underscores the excellent compatibility of LiFPB-containing electrolytes with Li metal anodes. Morphological characterization of lithium electrodeposits in Li||Cu cells provided mechanistic insights into

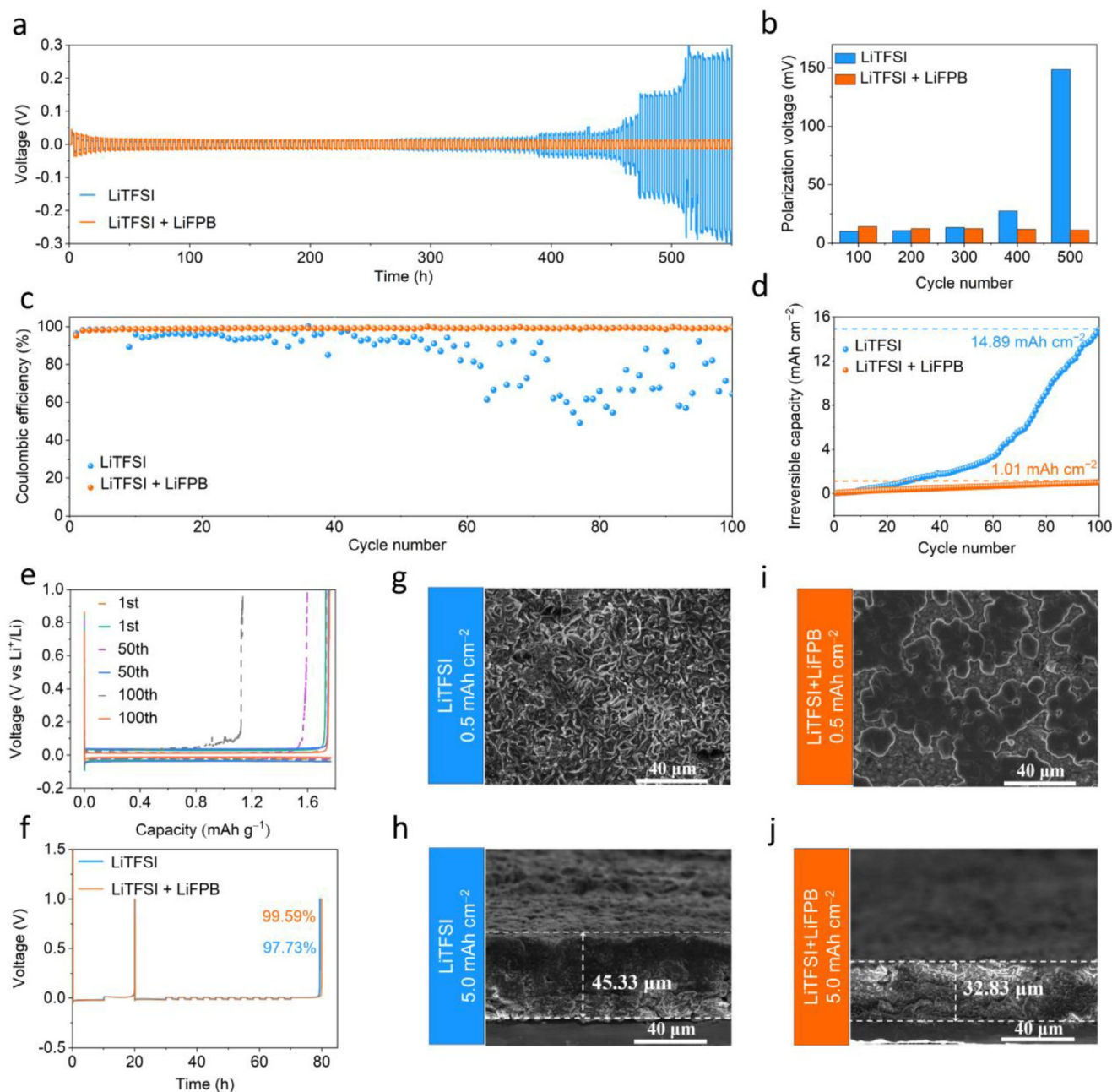


Figure 2. a) Voltage profiles of Li||Li symmetrical cells using different electrolytes at 0.5 mA cm^{-2} for 1 mAh cm^{-2} . b) Polarization voltages at different times in Li||Li symmetric cells with varied electrolytes. c) Coulombic efficiency of Li||Cu cells using varied electrolytes at 0.5 mA cm^{-2} for 1 mAh cm^{-2} . d) The cumulative irreversible capacity within the first 100 cycles in the varied electrolytes. e) Voltage profiles of the different cycles for Li||Cu cells using the varied electrolytes (dotted line: conventional electrolyte, solid line: LiFPB-containing electrolyte). f) Aurbach measurements in Li||Cu half cells. Typical SEM morphology of top-view and cross-section of Li deposition cycling after the first cycle at varied areal capacity in varied electrolytes: g, h) Conventional electrolyte. i, j) LiFPB-containing electrolyte.

the advantageous properties of LiFPB. At a controlled deposition capacity of 0.5 mAh cm^{-2} , conventional electrolyte formulations yielded heterogeneous, high-porosity, nodular lithium deposits, substantially amplifying the electrochemically active interface between lithium metal and electrolyte (Figure 2g,i). In stark contrast, the LiFPB-containing electrolyte facilitated the formation of large, densely-packed lithium particles on

the copper substrate, maintaining this favorable morphology even at escalated deposition capacities (Figure S7, Supporting Information).^[18,19] This favorable morphology effectively mitigated parasitic reactions and the formation of Li dendrites (Figure S8, Supporting Information). Additionally, the Li deposition thickness with the LiFPB-containing electrolyte was measured at $32.83 \mu\text{m}$, much thinner than $45.33 \mu\text{m}$ in

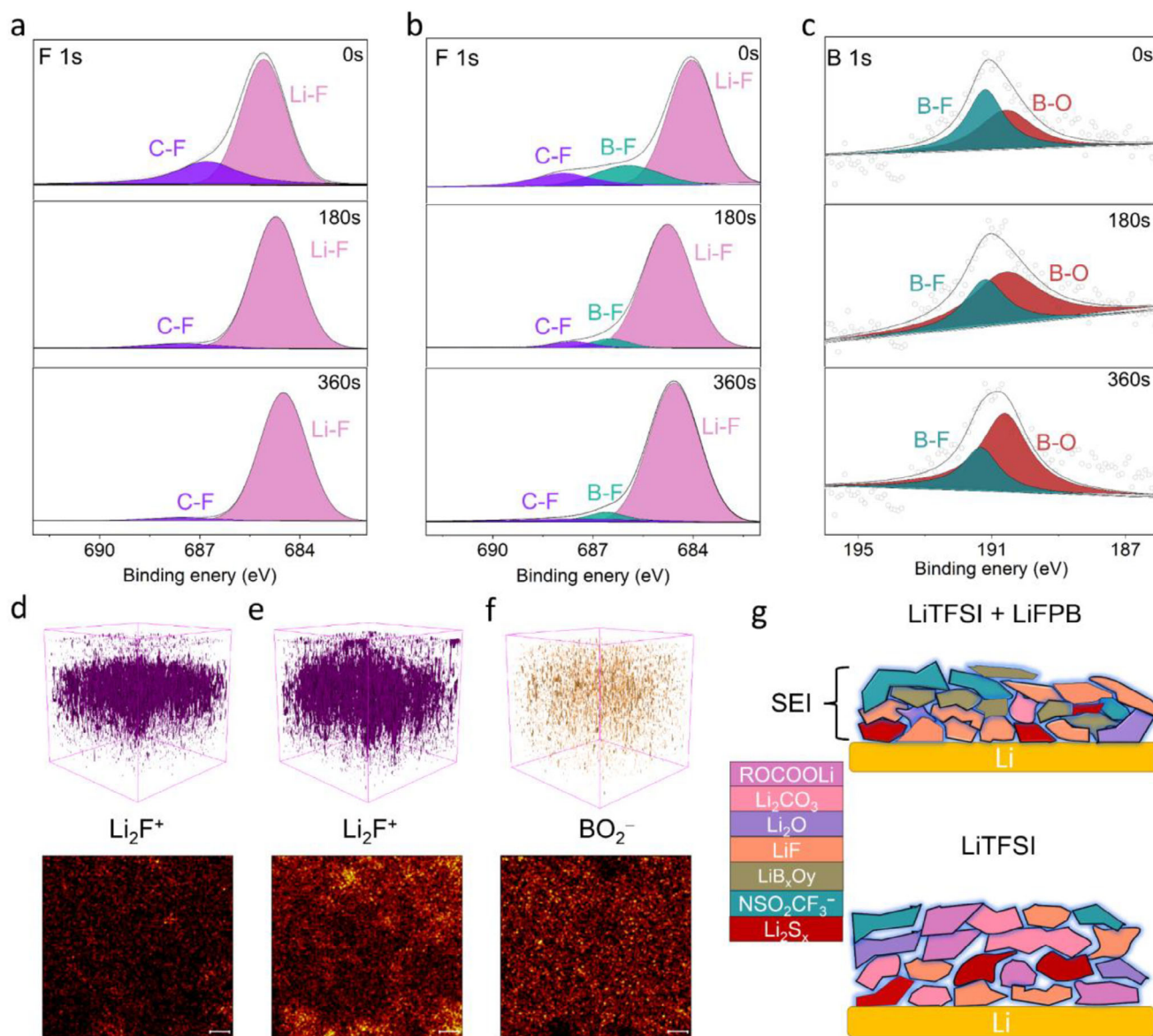


Figure 3. The XPS of Li deposition after 5 cycles in varied electrolytes: a) The XPS depth profiles of F 1s in the conventional electrolyte. b) The XPS depth profiles of F 1s in the LiFPB-containing electrolyte. c) The XPS depth profiles of B 1s in the LiFPB-containing electrolyte. TOF-SIMS 3D distributions of Li_2F^+ and corresponding mapping images in varied electrolytes: d) Conventional electrolyte. e) LiFPB-containing electrolyte. f) TOF-SIMS 3D distributions of BO_2^- and corresponding mapping images in LiFPB-containing electrolyte. g) Schematic illustration of component distribution in SEI that form under different electrolytes.

conventional electrolyte (Figure 2h,j). This result demonstrates that the LiFPB salt promotes uniform and dense lithium deposition, reduces micro-short circuits caused by SEI cracking, and limits dead Li formation. Consequently, LiFPB incorporation substantially enhances the electrochemical stability and compatibility of Li-metal batteries.^[20,21]

X-ray photoelectron spectroscopy (XPS) depth profiling was performed to elucidate the mechanisms underlying the enhanced cycling stability of Li metal in the presence of LiFPB salt. The Li metal anode cycled in LiFPB-containing electrolyte demonstrated substantially elevated fluorine-containing species

intensity compared to those cycled in conventional electrolyte (Figure 3a,b). The coexistence of B–F bonds with other inorganic components (such as LiF and Li_2O) facilitates the development of a moderately stable structure with well-defined lithium-ion conduction pathways. Survey spectra (Figure S9, Supporting Information) reveal a peak at ≈ 688.3 eV, corresponding to C–F bonds, which arise from intermediate products formed during the decomposition of LiFPB and LiTFSI on the lithium surface. Additionally, LiF (≈ 684.7 and ≈ 100.5 eV) was consistently detected throughout the sputtering depth. Notably, within the LiFPB-containing electrolyte system, the LiF signal intensity maintained remarkable stability with progressive etching depth,

whereas pronounced attenuation was observed in the conventional electrolyte.

Detailed spectroscopic analysis of the B 1s core-level spectrum (Figure 3c) confirmed the formation of boron-oxygen (B–O) coordination structures, which serve as strategically positioned electrophilic sites for preferential coordination with TFSI[−] anions. The selective anion immobilization mechanism effectively mitigates electrolyte decomposition through the formation of a robust anode-electrolyte interface.^[22,23] Comparative analysis revealed that lithium metal electrodes cycled in conventional electrolytes demonstrated progressive interfacial degradation, characterized by the time-dependent accumulation of electrolyte decomposition byproducts. X-ray photoelectron spectroscopy depth profiling identified the formation of organic carbonyl (C=O) and alkoxy (C–O) species, along with inorganic lithium carbonate (Li₂CO₃) compounds, whose concentration gradients exhibited a strong correlation with sputtering duration (Figure S10, Supporting Information). Depth profiling by time-of-flight secondary ion mass spectrometry (TOF-SIMS) corroborates these findings. In our TOF-SIMS analysis, a substantial signal for Li₂F⁺ was detected. Li₂F⁺ was commonly employed to identify the presence of LiF, thereby indicating that the electrolyte containing LiFPB is conducive to the formation of an SEI layer enriched with LiF. Notably, the LiF distribution exhibited both enhanced concentration and improved spatial homogeneity (Figure 3e) across the electrode interface, with these beneficial characteristics being maintained throughout the etching profile (Figure S11, Supporting Information). Spectroscopic and electrochemical analyses revealed that LiFPB facilitate the formation of lithium borate (LiB_xO_y) complexes at the lithium metal interface. These borate-containing species serve as rapid lithium-ion conductors within the solid electrolyte interphase (SEI),^[24–27] facilitating the uniform deposition of Li metal (Figure 3f). Depth-resolved chemical characterization demonstrated significantly enhanced retention of intact TFSI[−] anions at the electrode interface in LiFPB-containing electrolytes (Figure S11, Supporting Information),^[25] indicative of suppressed electrolyte decomposition pathways. This electrochemical stabilization originates from the formation of an ionically conductive and mechanically robust SEI layer, which effectively reduces the direct contact between the electrode and the electrolyte and minimizes parasitic reactions at the electrode-electrolyte interface (Figures S12,S13, Supporting Information).

Conventional electrolytes typically induce severe interfacial side reactions through electrolyte decomposition, resulting in the formation of a thick, heterogeneous, and porous solid electrolyte interphase (SEI). In marked contrast, our systematic characterization reveals that LiFPB incorporation enables the formation of an ultrathin, densely packed bilayer SEI architecture. This optimized interfacial structure comprises an outer organic-rich matrix and an inner inorganic-rich domain with substantial LiF enrichment (Figure 3g; Figure S14, Supporting Information). The rationally designed SEI configuration not only enhances the mechanical integrity and electrochemical stability of the lithium metal anode but also effectively suppresses parasitic reactions at the electrode-electrolyte interface. Consequently, the LiFPB-containing system demonstrates significantly improved cycling stability and rate capability.^[28]

To systematically assess the redox stability of the battery incorporating LiFPB, the exchange current density (I_0) was utilized as a quantitative metric for surface diffusion resistance (Figure 4a). Upon introduction of LiFPB into the electrolyte, the exchange current density associated with the electroplating and stripping of lithium metal exhibited a pronounced increase, escalating from 0.689 to 1.001 mA/cm², thereby representing a substantial 45.3% enhancement. This remarkable augmentation in interfacial charge transfer efficiency is principally attributable to the formation of a highly conductive solid electrolyte interphase (SEI) layer, which is characterized by optimized lithium-ion diffusion pathways. The presence of LiFPB in the electrolyte promotes the formation of an SEI layer that is rich in LiF and LiB_xO_y compounds. These compounds facilitate rapid lithium-ion transport, thereby significantly enhancing the overall conductivity of the SEI layer. The engineered anode interface architecture significantly contributes to the enhanced electrochemical performance and cycling stability of the battery system. Cyclic voltammetry (CV) of Li||Cu half-cells (0.2 mV s^{−1}) revealed superior redox characteristics in LiFPB-containing electrolytes, with higher peak currents and well-defined redox potentials versus conventional electrolytes (Figure 4b). This trend was confirmed in Li-S full cells across multiple scan rates (Figure 4c; Figure S15, Supporting Information).^[27] Furthermore, our investigations have elucidated that, in contrast to LiTFSI, LiFPB salt manifests a preferential reduction potential and a conspicuously higher reduction peak intensity (Figure S16, Supporting Information). This observation intimates that LiFPB salt is more susceptible to decomposition reactions and exhibits superior compatibility with metallic lithium, a propensity that can be attributed to the highly efficacious SEI layer it engenders on the negative electrode surface. Despite both salts being classified as readily reducible lithium salts, the SEI precipitated from the reduction of LiFPB markedly outperforms that of LiTFSI in terms of efficacy. Notably, the adsorption energy of LiFPB on the lithium metal surface is of a value higher than that of DME and DOL (Figure S17, Supporting Information). This finding unequivocally suggests that LiFPB exhibits a preferential adsorption affinity on the lithium metal surface.

The solvation structure was further analyzed by Raman spectroscopy. The LiFPB salt did not significantly affect the Raman shifts of DME, DOL, and LiTFSI (Figure 4d; Figure S18, Supporting Information),^[29] indicating that the SEI modification is not attributed to changes in the solvation structure. Electrochemical impedance spectroscopy (EIS) analyses performed on lithium symmetric cells (Li||Li) both prior to and subsequent to polarization with a 10 mV direct current (DC) unambiguously corroborated the salient benefits conferred by the incorporation of LiFPB salt. The impedance of the electrolyte containing LiFPB is significantly lower than that of the conventional electrolyte (Figure 4e), indicating that the SEI film formed by LiFPB salt can effectively facilitate the migration of lithium ions within the solid electrolyte interphase (SEI) and significantly reduce the interfacial resistance.^[30] The impedance measurements of Li||Li symmetric cells conducted at varying temperatures demonstrate that the LiFPB-containing electrolyte exhibits a significantly lower value compared to conventional electrolytes (Figure S19, Supporting Information). This reduction in impedance indicates that the LiFPB-containing electrolyte effectively lowers the internal resistance of the battery (Figure 4f).

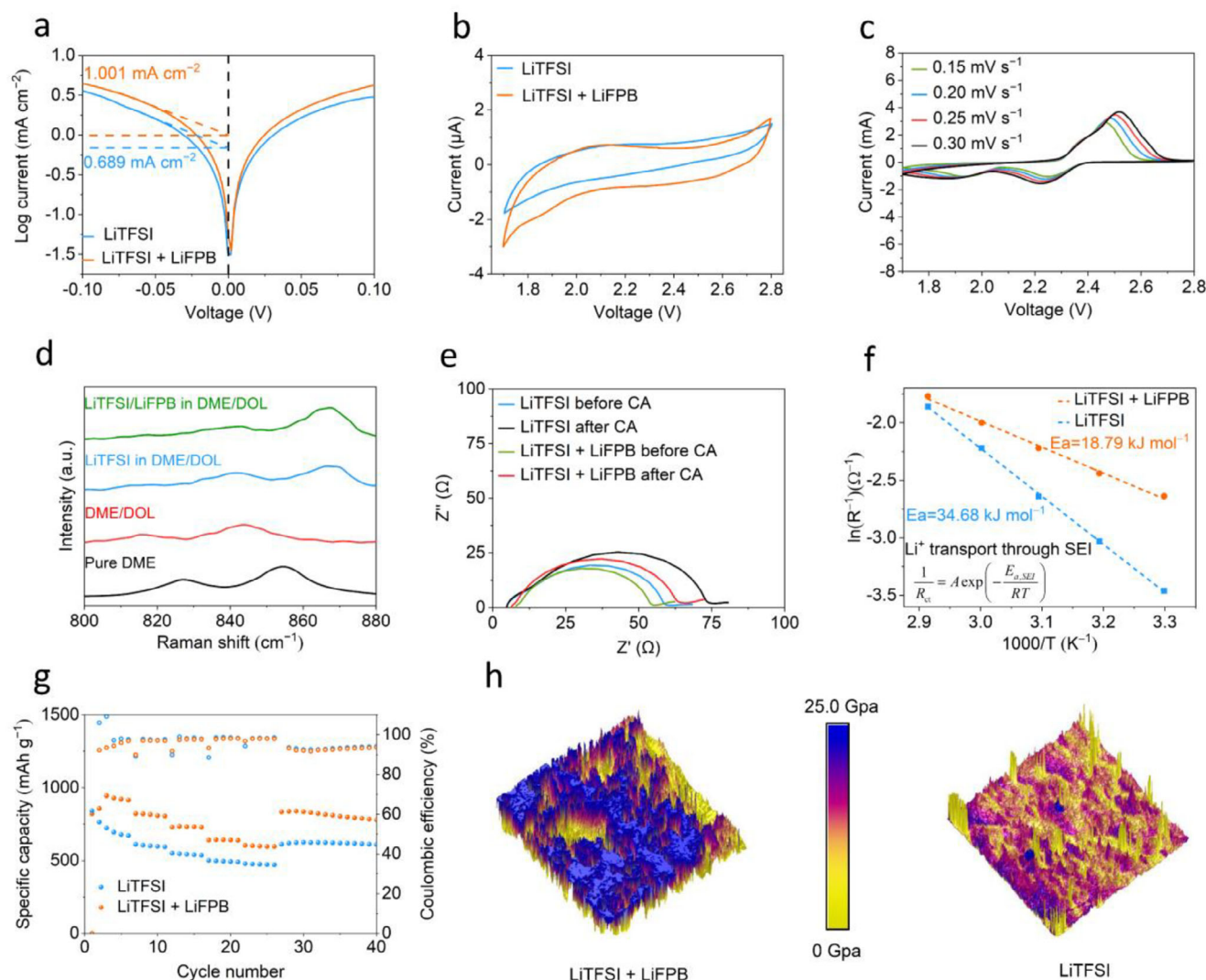


Figure 4. a) Tafel curves comparison of Li||Li symmetric cells employing varied electrolytes. b) Cyclic voltammogram curves (CV) of Li||Cu cell in conventional electrolyte and LiFPB-containing electrolyte with a scan rate of 0.2 mV S⁻¹. c) CV of Li||S cell in LiFPB-containing electrolyte with different scanning rates. d) Raman spectra of free DME, DME/DOL, LiTFSI-DME/DOL, and LiTFSI + LiFPB-DME/DOL. e) Electrochemical impedance spectroscopy (EIS) spectra before and after DC polarization in varied electrolytes. f) Li⁺ transport activation energies through SEI fitted by Arrhenius equation. g) Rate performance of Li-S batteries with varied electrolytes. h) Atomic Force Microscopy (AFM) images of the Young's modulus comparison of Li metal anodes after cycling.

Li-S batteries employing LiFPB-containing electrolytes exhibited exceptional rate performance (Figure 4g). The system delivered discharge capacities of 914.9, 805.8, 728.1, 638.7, and 596 mAh g⁻¹ at current densities of 0.1, 0.2, 0.4, 0.8, and 1.0 C. Remarkably, the capacity retention reached 837.6 mAh g⁻¹ upon returning to 0.1 C, surpassing the performance of conventional electrolytes. This enhanced electrochemical behavior can be attributed to the formation of a stable and ionically conductive SEI layer, which significantly improves cycling stability and redox kinetics.^[30] To quantitatively assess the mechanical properties of the SEI, we performed atomic force microscopy (AFM) Young's modulus measurements on cycled Li metal anodes from Li||Li symmetric cells (10 cycles, Figure 4h). The SEI derived from LiFPB-containing electrolyte demonstrated superior mechanical robustness, with a Young's modulus of 18.5 GPa, representing

a 3.3-fold enhancement compared to the conventional electrolyte system (5.6 GPa). After 20 cycles, the Young's modulus of the SEI layer in the electrolyte containing LiFPB was 24.6 GPa, compared to 10.8 GPa for the conventional electrolyte (Figure S20, Supporting Information). Surface topography analysis revealed that the LiFPB-modified anode exhibited reduced surface roughness (807.3 nm) compared to the LiTFSI-containing system (886.9 nm, Figure S21, Supporting Information). These observations collectively indicate the formation of a thinner, more homogeneous SEI layer with minimized parasitic reactions, highlighting the critical role of LiFPB in stabilizing the Li metal anode interface.

To assess the efficacy of LiFPB for lithium-sulfur (Li-S) batteries, cycling performance was evaluated under various rates. Following three activation cycles at 0.1 C, the batteries underwent

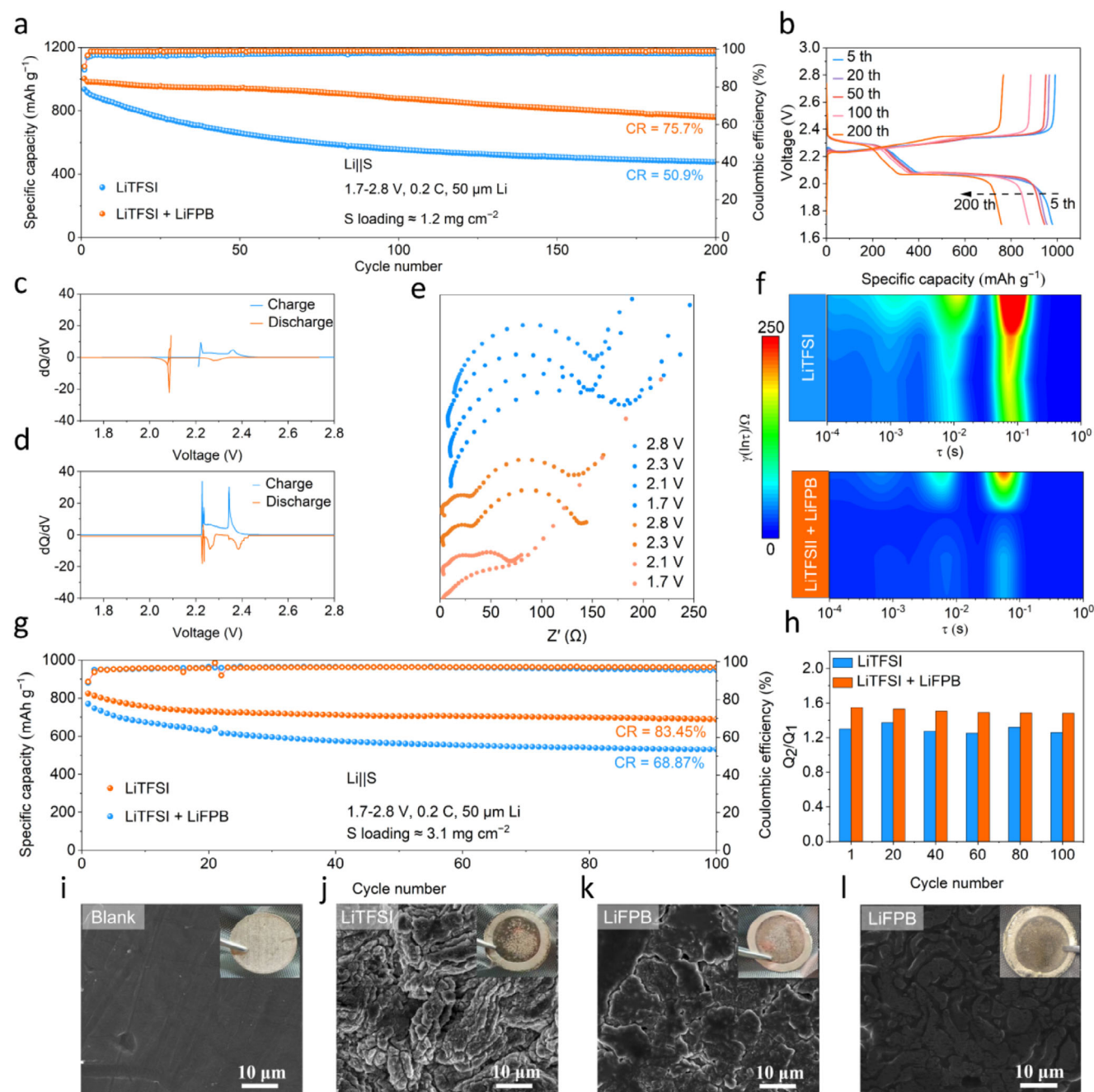


Figure 5. a) Cycling performance of Li-S batteries with S loading of 1.2 mg cm⁻², cycled at 0.1 C for the first three cycles and 0.2 C for the following cycles. b) Corresponding charge/discharge curves of Li-S batteries using LiFPB-containing electrolyte. c) dQ/dV curves of Li-S batteries after 100 cycles in varied electrolytes: c) Conventional electrolyte. d) LiFPB-containing electrolyte. e) In situ electrochemical impedance spectroscopy (EIS) spectra of Li-S batteries measured at various voltages in varied electrolytes. The corresponding DRT $\gamma(\tau)$ of in situ EIS in varied electrolytes. f) Corresponding capacity ratio of the second discharge plateau to the first discharge plateau (Q_2/Q_1) at different cycles. g) Cycling performance of the Li-S batteries with S loading of 3.1 mg cm⁻², cycled at 0.1 C for the first three cycles and 0.2 C for the following cycles. h) Corresponding capacity ratio of the second discharge plateau to the first discharge plateau (Q_2/Q_1) at different cycles. i) SEM morphology of i) Fresh Li metal, j) Typical cycled Li metal after 40 cycles in conventional electrolyte, k) Typical cycled Li metal after 40 cycles in the LiFPB-containing electrolyte, l) Typical cycled Li metal after 100 cycles in the LiFPB-containing electrolyte.

200 cycles at 0.2 C (Figure 5a). In the comprehensive evaluations of long-cycle stability and ionic conductivity across a range of LiFPB salt concentrations (Figure S22, Supporting Information) and other boron-containing salts (Figure S23, Supporting Information), it is noteworthy that the electrolyte system incor-

porating 0.2 M LiFPB salt demonstrated the most pronounced performance enhancement in long-cycle stability, thereby highlighting its superior electrochemical characteristics. The cell with the LiFPB-containing electrolyte exhibited a capacity retention of 75.7% (758.5 mAh g⁻¹/ 1001.8 mAh g⁻¹) after 200 cycles,

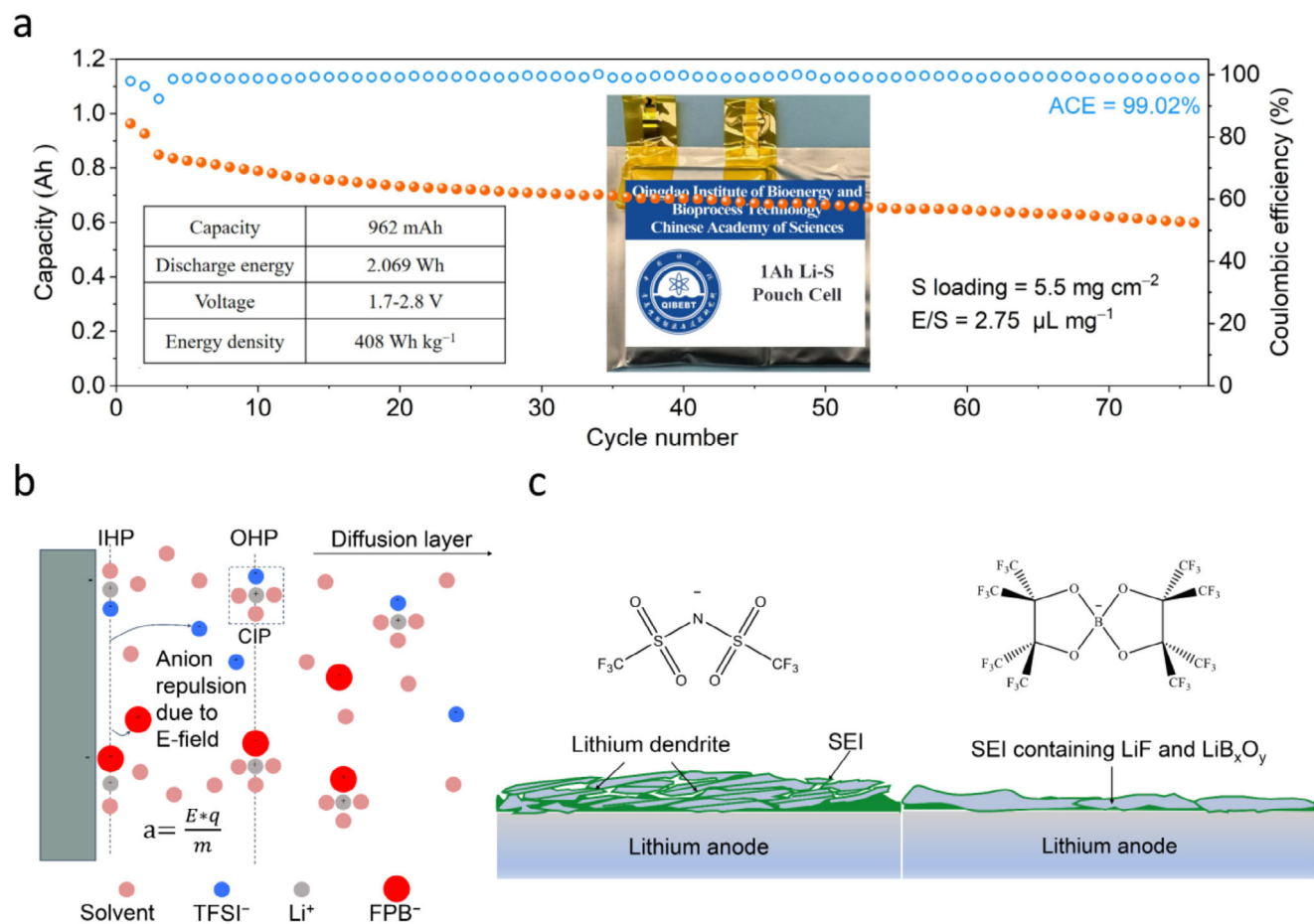


Figure 6. a) The specific energy and cycling stability of Li-S pouch cells. b) Schematic diagram of anion repulsion in the inner-Helmholtz layer due to the applied electric field. c) Schematic illustration of the positive role of LiFPB salt in protecting the lithium anode.

significantly surpassing the 50.9% retention observed for the conventional electrolyte (478.5 mAh g⁻¹/935.9 mAh g⁻¹) (Figure 5b; Figure S24, Supporting Information). Furthermore, the LiFPB-containing electrolyte achieved a higher average coulombic efficiency of 99.01%, compared to 97.62% in the conventional electrolyte. Similar trends were observed when the batteries were cycled at 0.5 C after three activation cycles at 0.1 C (Figure S25, Supporting Information), where the LiFPB-containing electrolyte demonstrated a capacity retention of 73.6%, notably outperforming the 56.41% retention of the conventional electrolyte. The average coulombic efficiency was also enhanced, increasing from 97.88% for the conventional electrolyte to 98.77% for the LiFPB-containing electrolyte. These findings highlight the significant improvements in cycling stability and coulombic efficiency that can be achieved by incorporating LiFPB under varying operational conditions.^[31,32]

To better understand the electrochemical behavior of lithium-sulfur batteries during cycling, differential capacity (dQ/dV) analyses after 100 cycles revealed that modified electrolytes raised the plateau voltages, reducing cell polarization (Figure 5c,d). This improvement is attributed to a better conductive network and reduced interfacial impedance, mitigating capacity loss and enhancing polysulfide deposition/dissolution uniformity while

suppressing the shuttle effect. In situ impedance and DRT analyses showed lower impedance for the modified electrolyte across all voltage ranges (Figure 5e). The stable SEI film enriched with LiF and LiB_xO_y reduces the charge transfer resistance in the high-frequency region (intrinsic electrolyte conductivity and interfacial contact), while the shortened lithium-ion diffusion pathways lower the impedance in the mid-frequency region (charging and discharging processes of the interfacial electric double layer). In the low-frequency region (liquid-phase diffusion of soluble lithium polysulfides), the mitigated diffusion polarization enables faster and more uniform lithium-ion transport (Figure 5f). These improvements, facilitated by the incorporation of LiFPB, enhance the stability of the solid electrolyte interphase (SEI), alleviate the shuttle effect, and significantly improve the utilization rate of the sulfur cathode and the cycling stability of the battery.

To assess the performance of the electrolyte, a high-loading lithium-sulfur (Li-S) battery with a sulfur loading of 3.1 mg cm⁻² was assembled and subjected to electrochemical evaluation. The battery utilizing LiFPB-containing electrolyte exhibited an initial specific capacity of 824.6 mAh g⁻¹ (Figure 5g). After 100 cycles, it retained a capacity of 688.1 mAh g⁻¹, corresponding to a capacity retention rate of 83.4%, with an average coulombic

efficiency of 97.96%. In contrast, the battery with the conventional electrolyte retained only 530.6 mAh g⁻¹ after 100 cycles, with an average coulombic efficiency of 96.71%, which is significantly lower than that achieved with the LiFPB-containing electrolyte. Furthermore, the discharge capacity ratio (Q_2/Q_1) between the two discharge plateaus of the Li-S battery, as assessed during long-cycle tests at 0.2 C, 0.5 C, and high-sulfur-loading conditions (Figure 5h; Figure S26, Supporting Information), provides a clear indication of the enhanced electrochemical kinetics of the sulfur electrode within the modified electrolyte. The capacity degradation during the aging process was effectively mitigated, leading to improved capacity retention and higher coulombic efficiency in high-loading Li-S batteries.

To investigate the underlying mechanisms, the morphology of the Li metal after 40 cycles was examined (Figure 5i–k). The cycled Li metal in the conventional electrolyte displayed severe corrosion, characterized by numerous cracks and a highly porous structure. This degradation is primarily attributed to continuous reactions between Li metal and polysulfides in the electrolyte. In contrast, the Li surface cycled with the LiFPB-containing electrolyte exhibited larger particles with a more uniform, adherent, and smooth morphology, even as the number of cycles increases to 100 (Figure 5l).^[33] The surface degradation observed in conventional electrolytes was likely attributed to non-uniform electric currents caused by cracking and porosity, which result in irregular lithium plating and stripping. These issues ultimately lead to a reduction in the cycle life of lithium-sulfur (Li-S) batteries. Incorporating LiFPB into electrolyte effectively suppressed the shuttle effect and facilitates the formation of a thin and uniform SEI. These improvements mitigated undesirable side reactions, promoted stable lithium deposition, and significantly enhanced the cycling stability and overall performance of Li-S batteries.^[34,35]

To evaluate the practical viability of the modified electrolyte, an Ah-level Li-S pouch cell was fabricated. The cell delivered an impressive energy density of 408 Wh kg⁻¹ at 0.1 C, maintaining 62.24% capacity retention with a high average coulombic efficiency of 99.02% over 76 cycles (Figure 6a). This result indicate that the incorporation of LiFPB into electrolyte can effectively relieve the corrosion of lithium metal by polysulfide-containing electrolyte in pouch cells by forming a robust SEI.^[36] The electrostatic repulsion of conventional anions within the inner Helmholtz layer (IHP), arising from the negatively charged electrode surface, hinders the formation of crucial inorganic SEI components. However, LiFPB anions, possessing a higher specific charge (mass-to-charge ratio), demonstrate reduced susceptibility to the applied electric field and consequently greater resistance to IHP repulsion (Figure 6b). For the in situ infrared spectroscopy analysis, a comparative examination of the C-F peak variations in the LiFPB and LiTFSI systems revealed a distinct correlation between the detected C-F signal intensity and the adsorption/desorption dynamics of anions within the inner Helmholtz layer as charge-discharge cycling progressed (Figure S27, Supporting Information). Specifically, the characteristic absorption peak of LiFPB was consistently observed throughout the entire cycling process, with its intensity fluctuations being markedly smaller than that of TFSI⁻. This finding provides compelling evidence that FPB⁻ is indeed present within the electric double layer and exhibits superior stability during the charge-discharge process, thereby indicating that LiFPB possesses a

significantly enhanced retention rate within the double layer. The resultant thin, ionically conductive, and robust LiF-rich and LiB_xO_y-rich SEI layer effectively suppressed Li dendrite growth (Figure 6c), ultimately enhancing the cycling stability of Li-S pouch cells.

3. Conclusion

In this work, we demonstrated that LiFPB served as a highly effective co-salt for Li-S batteries by simultaneously addressing two critical challenges: the polysulfide shuttle effect and the Li metal anode degradation. Mechanistically, the incorporation of LiFPB facilitated the formation of a robust SEI layer enriched with LiF and LiB_xO_y components, which significantly suppressed electrolyte decomposition and enabled homogeneous Li plating/stripping. The exceptional electrochemical performance can be rationalized by the unique molecular features of LiFPB, specifically its high specific charge capacity and favorable reduction potential, which synergistically enhance SEI stability and minimize parasitic side reactions. The practical viability of this strategy is evidenced by the superior performance in both high-loading Li-S half-cells and Ah-level pouch cells across various C-rates, representing a significant advancement toward the commercialization of high-energy-density Li-S batteries.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

T.L., D., Q., G.L., and G.C. conceived the idea for this project. D.Q., Z.C., T.L. and Y.S., and Y.Y. prepared the materials and carried out electrochemical tests and characterizations. Z.L., C.L., J.M., S.Z., Z.L., C.G., S.K., and G.X. discussed together and proposed the concepts of this paper. All authors participated in manuscript preparation.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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macromolecular boron-based salt, interfacial chemistry, Li-S batteries

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